

Radioactivity

WHEN AN ATOM HAS AN UNSTABLE NUCLEUS, IT CAN BREAK APART, EMITTING HIGH-ENERGY PARTICLES AND RADIATION.

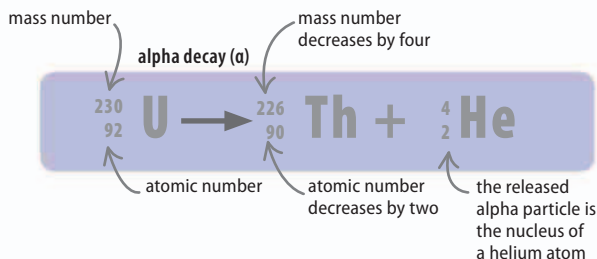
A radioactive atom is generally very large, and its nucleus has a different number of neutrons from a stable atom. It is called a radioactive isotope of the element.

Radioactive decay

When an unstable nucleus breaks apart, or decays, it produces radiation. Gamma rays are one type of radiation. They are the highest energy waves in the electromagnetic spectrum. Sometimes a nucleus will emit fast-moving particles. Losing these alters the structure of the nucleus and produces a new element. Alpha particles are made up of two protons and two neutrons—the same as the nucleus of a helium atom. Beta particles are generally single electrons.

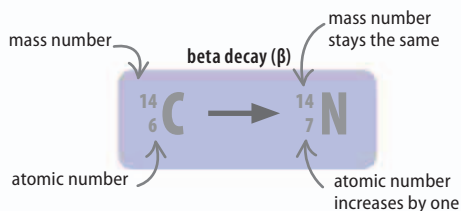
▽ Alpha decay

An alpha particle is formed when the parent atom's nucleus loses two protons and two neutrons. This decreases its atomic number (the number of protons in an atom) by two. So radioactive uranium (atomic number: 92) decays into thorium (90). The mass number (the number of protons and neutrons) decreases by four.



▽ Beta decay

A beta particle is formed when a neutron in the unstable nucleus splits into a proton and electron. The proton stays in the nucleus, raising the atomic number by 1, while the electron is pushed out. Therefore radioactive carbon atoms (atomic number: 6) form nitrogen (7). The nucleus has one less neutron but one more proton, so the mass number stays the same.



SEE ALSO

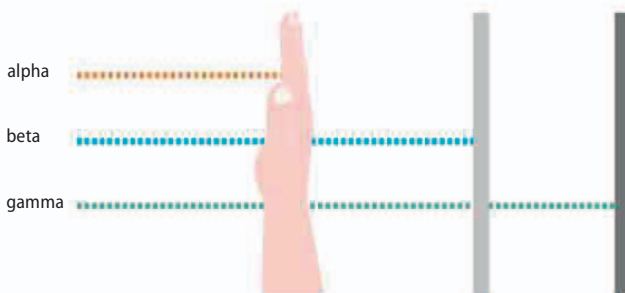
◀ 116–117	Periodic table
Inside atoms	168–169 ▶
Electromagnetic waves	194–195 ▶
Energy from atoms	219 ▶
The Sun	232–233 ▶

Dangerous radiation

Radioactive radiation is dangerous because it contains so much energy that it can ionize—knock electrons off—the atoms in living tissues. This damages the way cells work, causing them to die in large numbers, and can trigger cancers. Large alpha particles can only get into the body through food or drink, but can then cause a lot of damage. Gamma rays shine right through, but are less likely to hit a molecule and cause damage.

▽ Penetrating power

Alpha particles are blocked by the skin, although they can cause radiation burns. Beta particles bounce off thin sheets of metal, while it takes a thick layer of lead to shield against gamma rays.



REAL WORLD

Smoke detectors

Household smoke alarms contain tiny—and safe—amounts of americium, a radioactive element made in laboratories. The americium ionizes the air inside. A battery runs a current through it. When smoke gets in, the air is deionized and the current is blocked, triggering the alarm.

